Detecting and Mapping Memorization in ChatGPT Using Output Variance: A Case Study on Theoretical Data Structures Questions

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As large language models become increasingly common in computer science education, an important question is whether their provided solutions reflect genuine reasoning or retrieval of memorized logical patterns. This study presents a proof of concept pipeline for detecting and characterizing memorization behavior in theoretical Data Structures questions based on output-variance metrics. We significantly validate the link between low output variance and memorization ($p < 0.001$): we show that solutions for memorized questions tend to have higher semantic similarity, lower variance in reasoning tokens usage, and more stable grades. Clustering in the resulting metric space revealed a significant two-group structure ($p = 0.01$), confirming a meaningful distinction between memorization levels. Predictive models achieved performance clearly above chance when trying to predict the established clustering based on query content, with XGBoost reaching an AUC of 0.852, suggesting that memorization depends on structural combinations of topics and task formats. Overall, the results demonstrate that relatively simple output-variance metrics effectively identify memorization in complex reasoning tasks, and could use to analyze memorized topic distribution in closed-source language models.

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1 INTRODUCTION

As Large Language Models (LLMs) such as ChatGPT become widely used in Computer Science education, a key question arises: when these models solve logic-heavy problems, are they reasoning through the solution or retrieving patterns from their training data? Understanding this distinction is essential for evaluating the reliability of model outputs as well the causes or components of their establishment. This study explores this question in the domain of Data Structures, a discipline centered on algorithmic reasoning and computational complexity rather than the recall of prior knowledge.

In the context of complex subjects like Computer Science, memorization is more than just recalling simple facts. It involves the model reproducing specific algorithmic or computational ideas it has seen frequently during training. While we often think of memorization as a "yes or no" situation, in the domain of complex questions it could also be described as a spectrum where a model might synthesize parts of an answer while relying on memorized "templates" for others.

This intuition is related to prior work showing that memorized or contaminated examples tend to produce unusually confident or concentrated model behavior. When an LLM answers a genuinely new problem, there are often several valid ways to explain the solution, choose intermediate steps, or phrase the reasoning. In contrast, when a prompt activates a pattern that was strongly represented during training, the model may behave less variably: repeated generations are more likely to follow the same solution path and produce highly similar answers. This is consistent with work on memorization and training-data extraction (Carlini et al., 2021, 2022), output-distribution signals for contamination detection (Dong et al., 2024), and likelihood-based detection of pretraining data (Shi et al., 2024). Our work applies this intuition to complex Computer Science questions, where memorization may appear not as exact word-for-word reproduction, but as repeated reliance on the same reasoning template or algorithmic solution pattern.

In an era increasingly shaped by the risk of "cognitive surrender"—where the ease of AI-generated solutions may lead users to bypass critical thinking *(Shaw et al., 2026)*—protecting the integrity of the learning process is more important than ever. Detecting and analyzing memorization can support more critical use of AI by providing a diagnostic pipeline that helps identify instances of memorization and it's characteristics in the selected domain. By revealing what occurs "behind the scenes" of a generated response, this approach encourages users to distinguish between genuine synthesis (building an answer through reasoning) and rote retrieval (recalling a stored pattern).

Understanding the strengths and limitations of memorization in the Data Structures and Algorithms (DSA) domain can help students determine when it is appropriate to rely on language models while studying—whether for understanding lecture material, solving assignments, or preparing for exams. At the same time, it allows instructors to adapt teaching materials, homework, and exams according to how likely an LLM is to reproduce answers from memorized patterns, and to better assess the origin and quality of AI-assisted responses students may obtain.

Consequently, the goal of this project is to develop a proof of concept (POC) for a pipeline that identifies memorization in complex textual questions, focusing on Data Structures problems, while exploring factors that correlate with this behavior. Despite their relative simplicity, these output-variance metrics are tailored for black-box models used in educational contexts and still demonstrate high predictive power. Moreover, the ability to map "training intensity", as demonstrated in this POC, has implications in model assessment and training corpus characterization —- beyond computer science education.

The research methodology is structured as a systematic three-phase pipeline designed to identify and analyze memorization behavior, as visually outlined in Figure 1. This diagnostic workflow begins in Phase 1 with the establishment and validation of variance-based metrics using "likely memorized" and mixed exam datasets. These metrics serve as the foundation for Phase 2, where questions are projected into a normalized space for clustering-based classification. Finally, Phase 3 uses these classifications to identify the specific factors—such as course topics and task types—that either increase or decrease the likelihood of memorization.

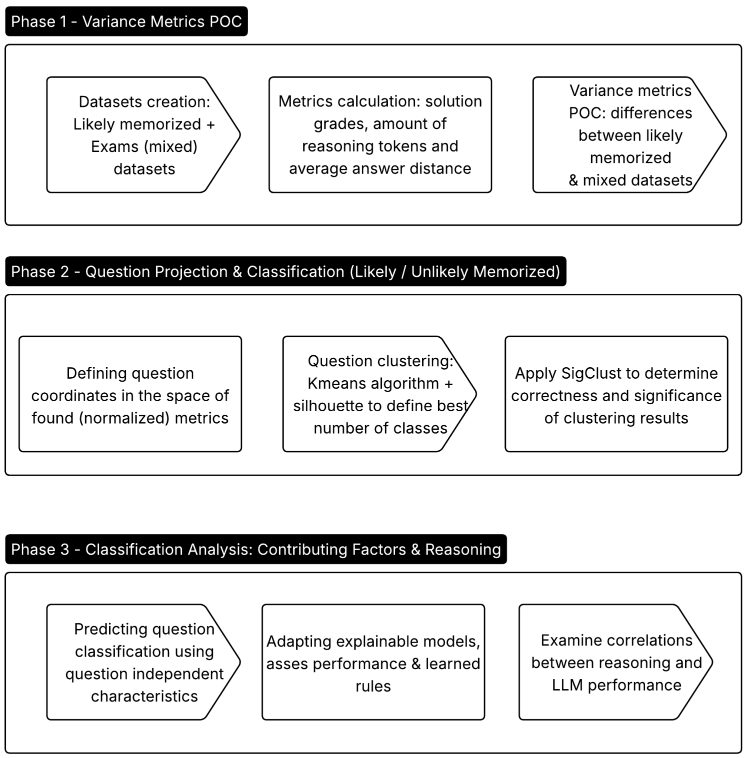

Figure 1. Three-Phase Research Pipeline: Provides an overview of the study's workflow: Phase 1 validates variance-based metrics; Phase 2 uses these metrics to cluster questions into "Likely" and "Unlikely" memorized categories; and Phase 3 analyzes the structural factors influencing this behavior.

2 RELATED WORK

Evaluation of large language models nowadays includes distinguishing genuine reasoning from rote memorization, beyond performance assessment.

A foundational branch of this research, led by Carlini et al., focuses on verbatim memorization—the model's ability to reproduce training sequences word-for-word. This work studies memorization through exposure-style measurements, training-data extraction attacks, and related membership-inference formulations. Carlini et al. (2021) rank generated samples using several membership-inference metrics, including model perplexity, ratios of log-perplexities between models of different sizes, and zlib-based ratios that compare model likelihood to the sequence's compressibility under a generic compression algorithm. Carlini et al. (2022) further quantify how extractable memorization scales with model size, data duplication, and prompt context length. Their work initially targeted GPT-2 trained on WebText, successfully extracting verbatim sequences like PII (names, phone numbers), IRC conversations, and 128-bit UUIDs even if they appeared only once in the training corpus.

The GPT-3 paper, "Language Models are Few-Shot Learners," identifies data contamination—the inadvertent inclusion of benchmark test sets in the training corpus—as a "major methodological concern". As models are trained on trillion-token datasets like Common Crawl, the likelihood of seeing evaluation questions beforehand increases. The GPT-3 authors developed tools to measure this, utilizing a 13-gram overlap analysis to flag potentially leaked examples. They defined "dirty" examples as those sharing at least 13 consecutive words with the training data.

These approaches provide important foundations for studying memorization and contamination, but they mainly target relatively explicit forms of overlap: either direct verbatim extraction, as in Carlini et al., or surface-level N-gram overlap, as in the GPT-3 contamination analysis. In contrast, the setting studied in our work involves complex logical questions, specifically in Data Structures and Algorithms (DSA), where memorization may be less explicit. A model may not reproduce a specific paragraph word-for-word, but may still follow a memorized solution pattern, proof structure, or standard reasoning flow encountered during training.

Dong et al. (2024) extend this line of work from direct extraction and overlap-based contamination checks toward detection based on the model's output behavior. They propose CDD, a method for detecting data contamination through the distributional patterns of generated outputs, based on the intuition that models tend to produce more concentrated and less variable responses for examples they have likely seen during training. This is especially relevant to black-box evaluation settings, since it does not require direct access to the training data. However, while CDD focuses on distributional signals for benchmark contamination, our work examines repeated answers to complex ("reasoning heavy") Data Structures and Algorithms questions and uses quality, semantic and behavioral variance–including answer-embedding dispersion, grade variance, and reasoning-token variance–as indicators of possible memorization.

Shi et al. (2024) address the related problem of detecting whether a given text was included in an LLM's pretraining data. They propose Min-K% Prob, a likelihood-based detection method relied on the intuition that unseen examples are more likely to contain a small fraction of unusually low-probability tokens, whereas seen examples tend to be assigned higher probability more consistently. While this approach is highly relevant to contamination detection, it relies on access to token probabilities; in contrast, our setting focuses on closed-source educational use cases where such probabilities are unavailable, and therefore uses variance across repeated generated answers as the main observable signal.

While the impact of contamination on performance is often debated, recent studies on reasoning benchmarks suggest that benchmark exposure can make models appear more capable than they are under clean evaluation conditions (Xie et al., 2024; Wu et al., 2025). Xie et al. study memorization in logical reasoning using dynamically generated Knights-and-Knaves puzzles, showing that fine-tuned models can nearly perfectly solve training puzzles yet struggle with slight perturbations, highlighting output performance variance caused by slight input changes as a sign of memorization. However, their results also suggest a nuanced interaction: memorization and improved generalization can coexist rather than being mutually exclusive. Wu et al. examine contamination in mathematical reasoning benchmarks and introduce RandomCalculation, a leakage-free arithmetic generator, showing that some apparent reinforcement-learning gains become unreliable once benchmark leakage is controlled.

These studies typically rely on perturbation tests or newly generated leakage-free benchmarks. In contrast, this work uses repeated black-box outputs on standard Data Structures and Algorithms questions and treats output variance as a proxy for memorization fingerprints. This direction is closest in spirit to output-distribution approaches such as CDD, but differs by criteria mentioned above. However, these works highlight the importance of memorization detection in logic, computational domains and the effect on model's approach to reasoning problems.

This motivates our use of output variance as a proxy for memorization. By analyzing the consistency of the model's behavior across multiple repeated generations, we aim to capture "memorization fingerprints" that may appear in reasoning even when no exact textual match is produced. This is particularly valuable for black-box, closed-source models used in education, where internal metrics such as logits or loss are usually unavailable. Furthermore, we show that question-level characteristics can predict strong memorization fingerprints with relatively high AUC, suggesting that output-variance metrics may help generate a "training intensity map": an estimate of which types of educational materials (or potentially other topics) are more densely represented, and possibly memorized, in the model's training corpus distribution.

3 PHASE 1: ESTABLISHMENT OF DATA AND OUTPUT-VARIANCE METRICS

3.1 Dataset Creation and Metric Calculation

3.1.1 Dataset Creation Pipeline. For the three phases of the study, two datasets were used.

The first dataset consists of exam questions from the Data Structures course at Tel Aviv University (TAU). This dataset includes extracted exam questions and solutions from the years 2022–2024, together with earlier TAU exam questions collected by prior students who took the workshop. In this dataset, the memorization status of questions is unknown. The assumption is that the dataset contains a mixture of questions that are likely memorized by language models and questions that are unlikely memorized, in an unknown proportion.

The second dataset was constructed to contain questions whose solutions are likely memorized by language models. Importantly, these questions are not necessarily easier for humans or shorter to solve. Rather, they correspond to arguments, proofs, or algorithmic constructions that appear frequently in common Data Structures learning materials. The dataset was initially created from questions written by Data Structures lecturers from the School of Computer Science at Tel Aviv University (Amir and Michal). Using these examples and additional instructions, the dataset was enriched with similar questions generated using GPT-5.2.

Unlike the exam dataset, these questions do not have official solutions. Therefore, reference solutions were produced using GPT-5.2 under a role-playing prompt intended to generate expert-level answers. This approach was considered reasonable because questions that are likely memorized typically correspond to well-known arguments or proofs whose correct structure is stable across sources.

For Phase 3, two attribute systems were constructed. The first describes the type of task required by the question (e.g., proof, algorithm design, complexity analysis), while the second is a hierarchical attribute system describing the conceptual elements from the course material required for solving the problem. The attribute schemes were initially generated using GPT-5.2. Task-type attributes were derived from the question text itself, whereas the hierarchical subject attributes were derived from the official solutions. Initial attribute sets were created based on small samples of questions and solutions. The attribute systems were subsequently refined in an additional iteration aimed at improving performance in Phase 3. This refinement used larger samples, incorporated Gemini (which showed better performance for this task), and leveraged a textual Hebrew summary of the 2022 course material.

All processes shown in Figure 2 were executed independently for each question. Each question was processed in a separate session. In particular, at no stage were both the question and an official solution provided in the same session.

Questions were translated from Hebrew to English with LaTex formatting using GPT-5. Each question was then solved 7 times independently using the o3 model. Each solution was graded using GPT-5 according to strict grading guidelines (prompt is included in the appendix), as it seemed that "model accountability" improved grading quality during the workshop's opening assignment. From these outputs several metrics were extracted: **(1)** The number of reasoning tokens used for solving each question was recorded, and the mean and standard deviation were computed across the seven solutions. **(2)** For each question, the mean and standard deviation of grades were computed as well.

(3) To measure semantic variation between solutions, each generated solution was embedded using the `text-embedding-3-large` model. Let $e_{i1}, \dots, e_{i7}$ denote the embeddings of the seven solutions for question i. The semantic variation of answers was computed as the average pairwise cosine distance between embeddings:

$$D_i = \frac{2}{k(k-1)} \sum_{1 \le j < l \le k} \left(1 - \frac{e_{ij} \cdot e_{il}}{\|e_{ij}\|_2 \, \|e_{il}\|_2}\right)$$

where $k = 7$.

Figure 2 presents the pipeline of dataset and metric creation described above.

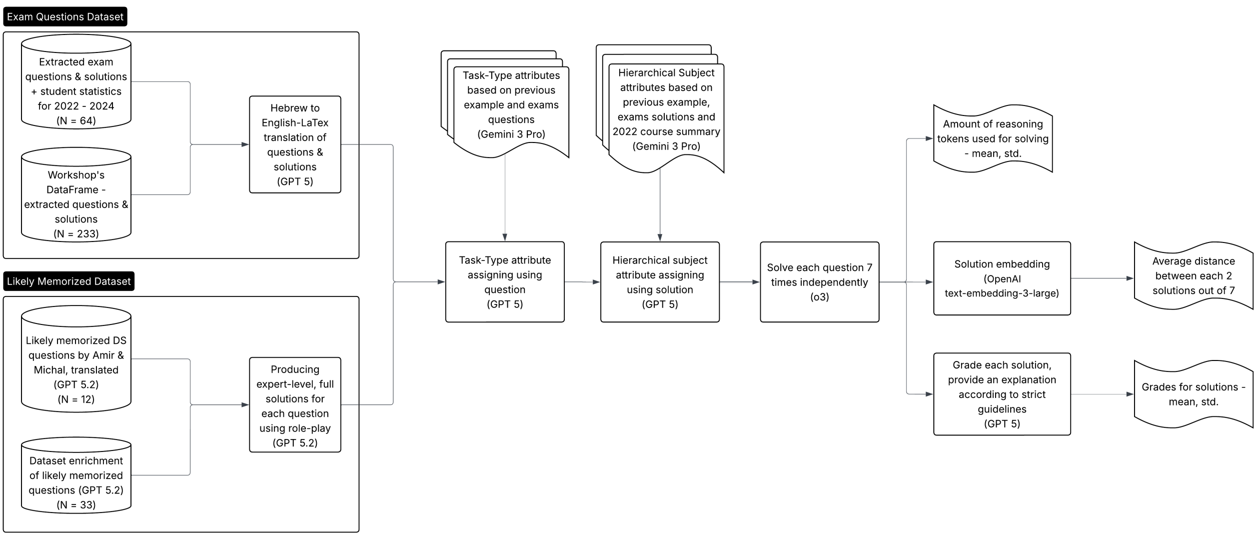

Figure 2. Dataset Construction and Metric Extraction : Illustrates the process of gathering exam and "likely memorized" datasets, translating them, and solving each question seven times to extract variance metrics such as grade stability, reasoning token count, and semantic embedding distance.

3.1.2 Metric Statistics. Table 1 presents descriptive statistics of the resulting metrics for each dataset. Variance metrics are consistently lower in the likely memorized dataset. This reduction across multiple dimensions provides the first quantitative evidence that the model produces significantly more stable outputs when retrieving familiar algorithmic patterns.

	avg_answer_distance		reasoning_tokens_mean		reasoning_tokens_std		grade_mean		grade_std	
	Exam	Likely Mem.	Exam	Likely Mem.	Exam	Likely Mem.	Exam	Likely Mem.	Exam	Likely Mem.
mean	0.182	0.106	1637.56	1459.61	735.99	403.19	75.07	89.37	13.91	5.37
std	0.081	0.039	1667.29	736.35	909.09	300.64	25.79	8.36	11.80	4.19
min	0.028	0.041	146.29	557.71	34.21	71.21	0.00	54.57	0.00	0.79
25%	0.132	0.077	603.43	932.57	212.26	195.02	61.86	87.57	3.29	1.98
50%	0.158	0.105	1152.00	1243.43	405.97	329.31	85.29	91.14	12.08	4.69
75%	0.205	0.133	1947.43	1865.14	811.11	426.82	95.00	93.86	21.44	7.20
max	0.545	0.202	11693.71	4041.14	5064.73	1442.89	100.00	99.29	48.60	16.03

Table 1. Summary Statistics of Calculated Metrics: Compares the distribution of metrics between the two datasets, showing that "Likely Memorized" questions consistently exhibit lower variance across different metrics and different quintiles of the question population, along with higher mean grades per question solutions.

Figure 3 illustrates the distribution of the analyzed metrics across the two datasets. To balance the sample sizes for this comparison, the exam dataset was bootstrapped 5 times with a sample size of 50; these samples were then sorted and averaged across the iterations for each metric. Similarly to Table 1, the variance-related statistics (standard deviations and average distance) are generally smaller for the likely memorized dataset than for the exam dataset (with the exception of the minimum values, which is expected given the larger original size of the exam dataset). In the next section, we formally test the significance of this observation.

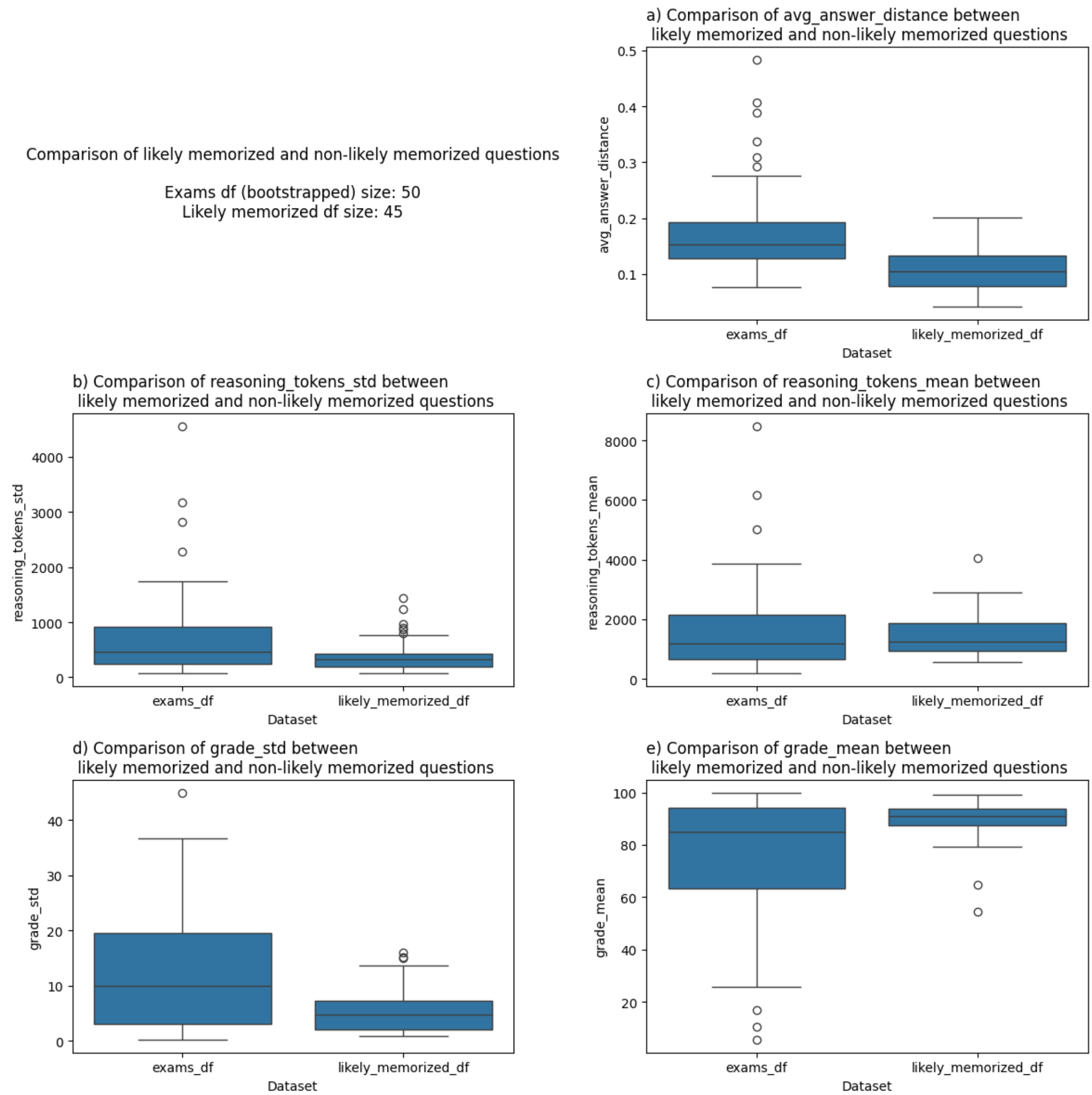

Figure 3. Comparison of Output-Variance, Grades and Reasoning Tokens Distributions: Box plots visually confirming that likely memorized questions (right) have tighter distributions and lower median values for semantic distance and reasoning token variance compared to the broader exam dataset. To balance sample sizes for comparison, the exam dataset was bootstrapped five times ($N = 50$), with metrics ranked and averaged across iterations to produce a representative distribution.

3.2 Proof of Concept for Output Variance Metrics

3.2.1 Statistical Analysis. After constructing the datasets—one consisting of questions that are likely memorized and another consisting of exam questions assumed to contain an unknown mixture of likely and unlikely memorized questions—we tested whether the output-variance metrics significantly differed between the datasets.

For each metric, we conducted an independent-samples t-test comparing the values computed on the TAU exam dataset and the "likely memorized" dataset. The tests were one-tailed, reflecting the directional hypothesis motivated by prior work (Carlini et al., 2021, 2022; Dong et al., 2024; Shi et al., 2024). Accordingly, the alternative hypothesis for each test was that the metric value is lower in the likely memorized dataset (grade mean is an exception).

Each metric was evaluated independently, as the purpose of Phase 1 is exploratory validation of candidate signals rather than joint hypothesis testing across metrics. The goal of these tests is to determine whether each metric individually shows evidence of distinguishing behavior between the two datasets. For this reason, we did not apply a multiple-comparison correction.

Importantly, these statistical tests are not intended to establish ground-truth memorization. Instead, they serve to validate the variance-based signals that are later used in Phase 2 for clustering analysis.

The results support the intuition suggested by the descriptive statistics (see table 2 below). Metrics capturing output variability—including the standard deviation of grades, the variance of reasoning tokens, and the average semantic distance between generated solutions—are significantly lower for the likely memorized dataset.

This finding is consistent with existing literature and suggests that when questions correspond to memorized material, language models tend to produce more consistent outputs across repeated attempts. Importantly, this observation holds even when the questions involve relatively complex reasoning, algorithmic arguments, and proofs from the domains of Data Structures and Algorithms. These results therefore provide preliminary validation both for the working assumption regarding the likely memorized dataset and for the output-variance metrics developed in this work.

It is worth noting that the comparison performed here is conservative. The exam dataset is assumed to contain an unknown mixture of likely memorized and unlikely memorized questions. If the comparison had instead been performed between the likely memorized dataset and a dataset consisting entirely of unlikely memorized questions, the differences between the groups would likely have been even more pronounced.

Interestingly, no statistically significant difference was observed in the mean number of reasoning tokens used in solutions across the two datasets. This does not necessarily imply that such a difference does not exist. The comparison is performed against a mixed dataset with unknown proportions of memorized questions, the number of reasoning tokens serves only as an indirect proxy for reasoning intensity, and other sources of noise may influence this metric. However, in contrast to prior work *(Wu et al., 2025)*, we do not rely on the assumption that reasoning and memorization are complementary mechanisms, or that questions prone to memorization necessarily require less reasoning on average.

Metric	H_0 (one-tailed)	t-statistic	p-value	Significant
Grade mean	$\mu_{\text{mem}} \geq \mu_{\text{exams}}$	-7.3436	0.0000	True
Grade std	$\mu_{\text{mem}} \leq \mu_{\text{exams}}$	-9.2125	0.0000	True
Average answer distance	$\mu_{\text{mem}} \leq \mu_{\text{exams}}$	-10.2349	0.0000	True
Reasoning tokens mean	$\mu_{\text{mem}} \leq \mu_{\text{exams}}$	-1.2162	0.1131	False
Reasoning tokens std	$\mu_{\text{mem}} \leq \mu_{\text{exams}}$	-4.8079	0.0000	True

Table 2. Statistical Significance of Variance Signals: Results of one-tailed t-tests showing that grade variance, semantic distance, and reasoning token variance are all significantly lower $(p < 0.001)$ for memorized content than mixed content, validating these as reliable signals for detection. Additionally, memorized content results in significantly higher average grades. While the mean number of reasoning tokens varies between groups, this difference is not statistically significant ($p = 0.1131$), suggesting that reasoning length is not a definitive indicator of memorization in this domain.

3.2.2 Manual Review. In addition to a manual review of 2–3 questions written by Amir and Michal (approximately 17% of the questions they contributed), a stratified sample of 10% of the exam-question dataset was manually inspected. The stratification was based on two variables: the classification of the question as likely or unlikely memorized (according to the results of Phase 2), and whether the question's average grade was above or below 70.

During the manual review, two aspects were examined. First, the correctness of the labels assigned to each question was evaluated. Second, the variability of generated solutions was inspected directly in the textual outputs. For each sampled question, approximately three solutions were manually reviewed out of the seven generated solutions.

The sampled questions and corresponding solutions are available in the Git repository referenced in the Appendix.

Label Quality. The manual inspection suggests that the attribute assigning process is generally reliable.

- In the reviewed sample, no cases were found where a clearly incorrect label was assigned to a question.
- However, the secondary task-type label was often left empty, even in cases where appropriate labels could plausibly have been assigned.
- In particular, many questions contained an element of complexity analysis, yet this aspect was frequently not captured by the labeling system, even as a secondary task type.
- A related issue arises from the hierarchical topic labeling scheme: in some cases the model labeled a specific subtopic but omitted its corresponding parent topic. This required handling during the analysis stage to ensure that when a subtopic appeared, the parent topic was also included.

Qualitative Analysis of Solution Variability. Manual inspection of the generated solutions revealed several types of variability patterns. These patterns differed between questions classified as likely memorized and those classified as unlikely memorized.

- **Variation in final answers.** In some cases, the final answer itself varied across generated solutions—for example, different choices in a multiple-choice question or different final mathematical expressions such as complexity bounds. Both the presence of disagreement and the distribution across alternative answers were substantially higher among questions classified as unlikely memorized.
- **Variation in justification.** Some solutions produced the same final answer but differed in the reasoning or justification used to reach it. Such patterns appeared in both groups, but were somewhat more common among questions classified as unlikely memorized.
- **Variation in textual structure.** Solutions sometimes expressed similar ideas using different textual structures. A highly consistent structure was a particularly strong characteristic of dominant likely memorized questions. This was especially evident in proof-style questions and in the questions written by Amir and Michal. In several cases within the likely memorized group, the wording itself was also strikingly similar across solutions.

Overall, the textual inspection confirms that memorization appears to operate along a spectrum. Even questions classified as likely memorized exhibited varying degrees of the patterns described above. Nevertheless, clear differences in the patterns above were observed between the two groups, particularly between the most representative likely memorized questions and those classified as unlikely memorized.

The manually reviewed examples can be found in the root directory of the attached Git repository. They are not included in this document due to their length, as each question contains multiple generated solutions.

4 PHASE 2: QUESTION PROJECTION AND CLASSIFICATION

4.1 Objective

The goal of this phase was to use the metrics that showed significant results in Phase 1 (grade standard deviation, reasoning tokens standard deviation, and average answer embedding distance) in order to characterize the behavior of different DSA questions and, if possible, classify them as likely or unlikely memorized.

4.2 Methods

To ensure that the relative scales of the metrics would not bias the analysis, each metric was normalized by subtracting its mean and dividing by its standard deviation.

Each metric then defined one axis in a three-dimensional space, resulting in a representation where every question corresponds to a point:

$$q_i = (d_i, r_i, g_i)$$

where d_i is the normalized average answer distance, r_i is the normalized reasoning tokens standard deviation, and g_i is the normalized grade standard deviation.

To explore whether meaningful structure exists in this space, clustering was performed using the k-means algorithm. Since the number of clusters k is a hyperparameter, we avoided selecting it manually and instead used the silhouette score to estimate the optimal number of clusters.

For a sample i, the silhouette value is defined as

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}$$

where $a(i)$ is the average distance between i and other points within the same cluster, and $b(i)$ is the smallest average distance between i and points in any other cluster. The silhouette score ranges between -1 and 1, with higher values indicating better cluster separation.

4.3 Clustering Results

Running the clustering algorithm on the combined datasets produced several notable results (Figure 4 below). First, silhouette analysis identified $k = 2$ as the optimal number of clusters , a structure that effectively aligns with the distinction between likely memorized and non-memorized question categories.

Second, the visual structure of the clusters is consistent with the expected behavior of likely and unlikely memorized questions clustering: The first cluster (yellow) is highly concentrated near the lower regions of all axes, corresponding to questions with consistently low output variance across metrics. In contrast, the second cluster (purple) is significantly more dispersed and extends toward higher values along the axes.

Third, all questions from the likely memorized dataset were classified into the first cluster (yellow), which we associated above with likely memorized questions. Moreover, within the likely memorized dataset, $2/3$ of the GPT-enriched questions and $5/6$ of Amir and Michal's questions had a **threefold smaller** distance from the centroid associated with likely memorized questions than from the centroid associated with unlikely memorized questions.

The resulting cluster centers were

$$C_0 = (1.0508,\ 1.3205,\ 0.8540)$$

$$C_1 = (-0.3261,\ -0.4098,\ -0.2650)$$

suggesting that one cluster corresponds to substantially higher variance values across all three dimensions.

4.4 Statistical Validation of Clustering

To determine whether the clustering reflects meaningful structure in the data rather than random variation, we applied the SigClust test, a statistical test specifically designed to evaluate the significance of clustering results in high-dimensional, low-sample settings *(Liu et al., 2008)*.

SigClust evaluates the null hypothesis that the data originate from a single Gaussian cluster with covariance matrix Σ:

$$H_0 : X_i \sim \mathcal{N}(\mu, \Sigma)$$

Under this hypothesis, cluster separation should arise only from Gaussian noise.

The test statistic is based on the within-cluster sum of squares:

$$W = \sum_{k=1}^{K} \sum_{x_i \in C_k} \|x_i - \bar{x}_k\|^2$$

where C_k denotes cluster k and $\bar{x}_k$ is its centroid.

SigClust estimates the distribution of W under the null model via simulation and computes the probability of observing a separation at least as strong as the one found in the data.

The resulting p-value was

$$p = 0.01$$

indicating that the two-cluster structure explains the data significantly better than a single cluster with Gaussian noise.

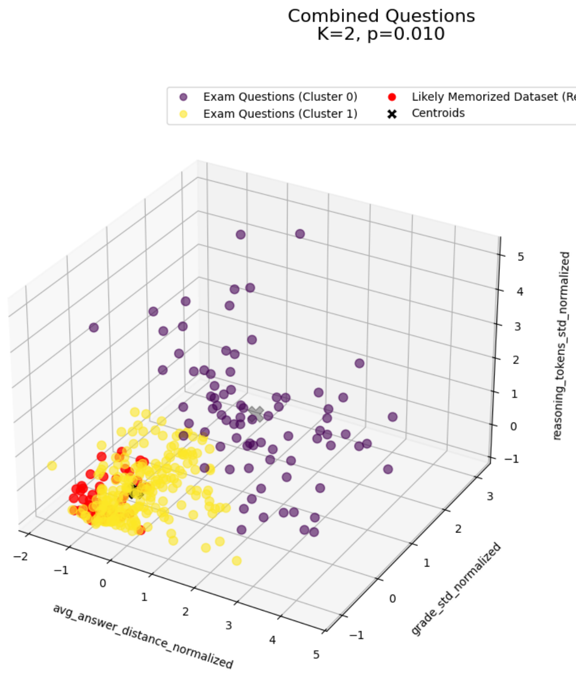

Figure 4. 3D K-Means Clustering of Questions : Visualizes the separation of the combined dataset into two distinct groups ($p = 0.01$): a highly concentrated "likely memorized" cluster (yellow) at the low end of all variance axes and a dispersed "unlikely memorized" cluster (purple). Questions from the "Likely Memorized" dataset are shown in red, consistently related to the memorized cluster.

4.5 Main Take for the Next Phase

These findings suggest that the discovered clusters capture meaningful structure in the question space and likely correspond, at least partially, to the distinction between likely memorized and unlikely memorized questions. Therefore, the clustering provides a potential basis for a meaningful classification of questions into likely and unlikely memorized categories in the next phase of the pipeline.

4.6 Prior Attempts

Several alternative approaches were also explored but yielded less informative results. In particular, we attempted Bayesian inference on the mixed exam dataset to estimate the memorization class of individual questions; however, the approach proved unstable due to the unknown mixture proportions.

We also experimented with DBSCAN clustering instead of k-means. In retrospect, the weaker performance of DBSCAN is unsurprising, since the clusters exhibit substantially different densities, whereas DBSCAN relies on a global distance threshold as its primary hyperparameter.

5 PHASE 3: CLASSIFICATION ANALYSIS

5.1 Objectives

The objective of this phase is to leverage the classification obtained in Phase 2 to identify explanatory factors that contribute to memorization behavior in model outputs. Rather than detecting memorization from the outputs themselves, this phase focuses on characterizing properties of the questions that increase the likelihood of memorization. In addition, we aim to further analyze the phenomenon by examining when memorization leads to low-quality answers, the relationship between reasoning length and performance, and the extent to which model behavior aligns with student performance.

5.2 Preprocessing

5.2.1 Label Refinement and Feature Encoding. Before training the classification models, the labeling systems described in Phase 1 were refined: The task-type attributes and hierarchical course-topic labels were encoded as one-hot features. The labeling process was further improved using the existing label format with a larger set of examples together and a textual summary of the Data Structures course material. Gemini was used in this step as it had better performance with longer course summaries. This refinement resulted in a modest performance improvement (approximately a 1% increase in AUC) and, based on manual inspection, produced labels that better reflected the structure of the course material and its different conceptual components.

5.2.2 Filtering Borderline Samples. Although Phase 2 demonstrated a proof of concept for separating likely and unlikely memorized questions using clustering, the resulting clusters do not guarantee perfect classification of the full question population. In particular, questions located near the boundary between clusters may be assigned an incorrect class.

To reduce the influence of such borderline samples, an additional filtering step was applied. For each question classified as likely memorized, we computed the ratio between its distances to the two cluster centers:

$$R_i = \frac{\mathrm{dist}(q_i, C_1)}{\mathrm{dist}(q_i, C_0)}$$

where C_0 denotes the centroid of the likely memorized cluster and C_1 denotes the centroid of the second cluster. To retain only high-confidence examples, we filtered the likely-memorized-classified questions and kept only questions with $R_i \leq 0.33$. This threshold ensures that the selected questions are at least three times closer to the likely memorized cluster than to the other cluster. The threshold was chosen based on visual inspection of the empirical distribution shown in Table 3.

Statistic	Value
count	261
mean	0.367
std	0.229
min	0.0669
25%	0.212
50%	0.276
75%	0.448
max	0.984

Table 3. Distance Ratio Statistics : Summarizes the proximity of questions to their respective cluster centroids; these ratios were used to filter "high-confidence" samples for the subsequent predictive analysis.

Figure 5 shows the resulting filtered dataset in the normalized metric space. As expected, the retained points form a tighter region around the low-variance cluster.

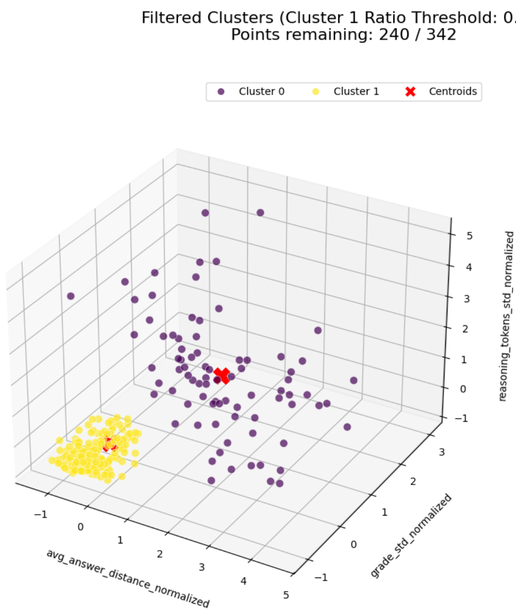

Figure 5. High-Confidence Filtered Clusters : Shows the dataset after removing borderline samples (those with a distance ratio > 0.33), resulting in more isolated and distinct clusters for cleaner feature analysis in Phase 3.

5.3 Classification Prediction: Logistic Regression

To predict whether a question is likely memorized, we trained a logistic regression classifier using the feature sets constructed in the preprocessing stage and the classes from phase 2 (after filtering). Logistic regression was chosen as a baseline model because it offers interpretability: model coefficients correspond directly to log-odds contributions of features.

5.3.1 Model Setup. The feature space contains a relatively large number of variables, particularly when interaction terms are included. Given the limited dataset size, this creates a risk of overfitting due to the curse of dimensionality. To mitigate this, the models were trained with LASSO regularization and evaluated using k-fold cross-validation.

LASSO adds an L_1 penalty to the logistic regression objective:

$$\min_{\beta} \left[-\sum_{i=1}^{n} \left(y_i \log p_i + (1 - y_i) \log(1 - p_i) \right) + \lambda \|\beta\|_1 \right]$$

where

$$p_i = \frac{1}{1 + e^{-\beta^T x_i}}$$

and λ controls the strength of regularization. The L_1 penalty encourages sparse solutions, effectively performing feature selection by shrinking many coefficients to zero.

Several model configurations were tested in order to limit feature dependencies and reduce dimensionality. In particular, the subject attributes were organized into two alternative hierarchical representations, and each model used only one hierarchy at a time. In addition, some experiments included student performance statistics associated with exam questions; however, this required restricting the dataset to a smaller subset containing those statistics.

Higher-order interaction terms were also explored. Third-order interactions were tested but did not improve performance and were therefore excluded from the final models.

5.3.2 Model Comparison. The best-performing models achieved a cross-validated ROC AUC of approximately 0.73. The ROC AUC measures the probability that the classifier ranks a randomly chosen positive example higher than a randomly chosen negative one. An AUC of 0.5 corresponds to random guessing, whereas 1.0 corresponds to perfect classification. Thus, while the model is far from perfect, its performance is significantly better than random and indicates that meaningful predictive patterns exist in the feature set. Table 4 summarizes the evaluated model configurations.

Model Name	Sample Size	Features In	Active After Lasso	CV ROC AUC	Accuracy	F1	FNR	FPR
TaskPri_SubPri_AllData_NoScores	208	33	60	0.729	0.841	0.872	0.118	0.222
TaskPri_SubSec_AllData_NoScores	208	117	95	0.727	0.904	0.922	0.071	0.136
TaskSec_SubPri_Subset_WithScores	47	35	27	0.684	0.936	0.955	0.059	0.077
TaskPri_SubPri_Subset_WithScores	47	35	31	0.679	0.936	0.957	0.029	0.154
TaskSec_SubPri_AllData_NoScores	208	33	96	0.659	0.851	0.877	0.126	0.185
TaskSec_SubSec_AllData_NoScores	208	117	118	0.649	0.889	0.906	0.126	0.086
TaskSec_SubSec_Subset_WithScores	47	119	39	0.599	0.915	0.944	0.000	0.308
TaskPri_SubSec_Subset_WithScores	47	119	0	0.574	0.277	0.000	1.000	0.000

Table 4. Logistic Regression Performance Comparison : Compares different model configurations, with the most effective setup (Task and Subject primary attributes) achieving a cross-validated ROC AUC of 0.729, significantly better than random guessing.

5.3.3 Feature Importance Analysis. To better understand which question characteristics influence memorization behavior, we analyzed the model coefficients and their interaction terms.

Because interaction features complicate interpretation, feature importance was computed by aggregating the contributions of all coefficients in which a base feature appears. For each base feature f, the total magnitude of influence was defined as

$$I_f = \sum_{j:f \in \phi_j} |\beta_j|$$

where ϕ_j represents a feature or interaction containing f. The direction of the effect was determined by the sign of the corresponding coefficient.

Algorithmically, this aggregation was implemented by summing the absolute coefficient magnitudes across all terms containing each base feature, while retaining the directional sign of the effect.

Figure 6 presents the aggregated log-odds impact of the most influential features. Several patterns emerge: topics such as asymptotic analysis, sorting, arrays and lists, hashing, and heap structures show strong positive contributions, indicating that questions involving these concepts are more likely to fall into the memorized category. These topics correspond to canonical algorithmic problems that frequently appear in textbooks and educational resources, making them plausible candidates for memorized patterns in language model training data.

Conversely, task types such as “Select Answer and Justify”, recurrence reasoning, and lower-bound proofs show negative contributions. These tasks often require contextual reasoning or justification rather than the application of standard algorithmic templates, which may explain their association with non-memorized behavior.

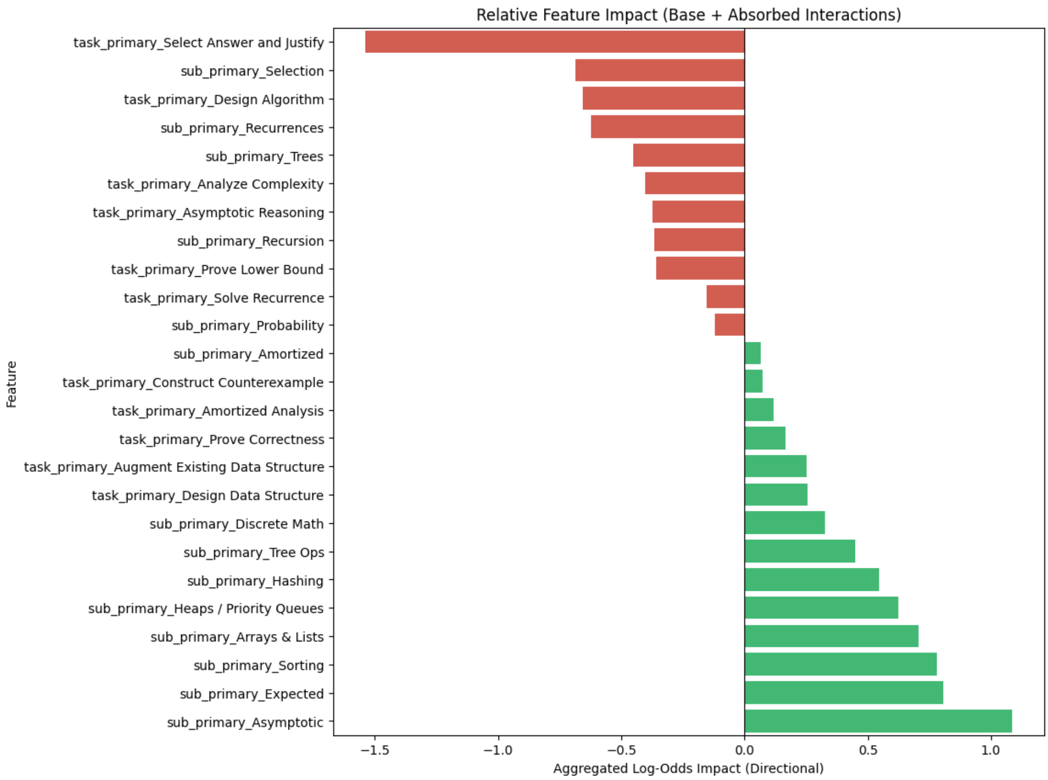

Figure 6. Feature Impact on Memorization Likelihood : Bar chart showing that canonical topics like Asymptotic analysis, Sorting, and Heaps are strong predictors of memorization, while task types requiring a choice from multiple options followed by explanation strongly predict non-memorized reasoning.

5.3.4 Direct vs. Interaction Effects. An important observation is that some features exert a stronger influence through interactions than through their direct coefficient. In several cases the interaction effect even reverses the direction of the direct effect. This suggests that memorization behavior cannot be explained solely by individual topic labels or task types; instead, it depends on the context given to such feature through combinations.

For example, certain topics may appear memorized only when paired with specific task types (e.g., algorithm design or asymptotic reasoning). Such contextual dependencies indicate that memorization patterns are partly structural and arise from typical pairings of concepts and problem formats in educational materials. Figure 7 compares the direct effect of individual features with their contextual influence through interactions with other features.

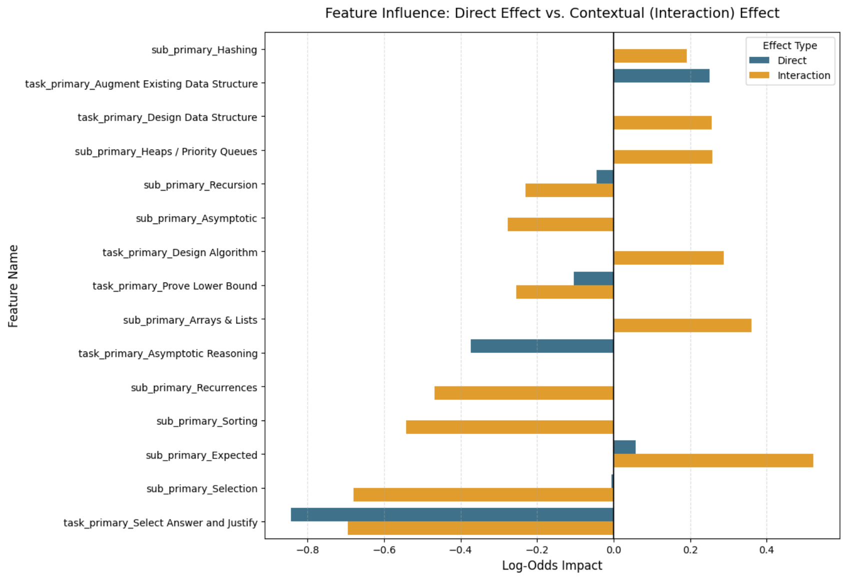

Figure 7. Direct vs. Contextual Feature Influence : Illustrates that for many topics, the interaction effect (orange) is more powerful than the direct effect (blue), suggesting that memorization often depends on specific combinations of topics and task formats.

Overall, the logistic regression analysis supports the hypothesis that memorization in complex questions is associated not only with specific topics but also with particular combinations of topics and task types.

5.4 Classification Prediction: XGBoost

To complement the interpretable logistic regression model, we trained a gradient boosted tree model using XGBoost. Tree-based models can capture nonlinear relationships and complex interactions between features that may not be easily expressed in linear models.

5.4.1 Model Configuration and Hyperparameter Search. XGBoost hyperparameters were selected using grid-search cross validation. The explored parameter grid was

$$\text{param\_grid} = \begin{cases} \text{interaction degree} \in \{1, 2\} \\ \text{max depth} \in \{2, 3\} \\ \text{n estimators} \in \{50, 100, 150\} \end{cases}$$

The best performing configuration was:

$$\text{Degree} = 1, \quad \text{Max Depth} = 2, \quad \text{N Estimators} = 100$$

With a learning rate of 0.05.The resulting cross-validated ROC AUC was $AUC_{validation} = 0.765$, and when evaluated across the full sample the model achieved $AUC_{full} = 0.852$.

Figure 8 shows the ROC curve for the optimized model.

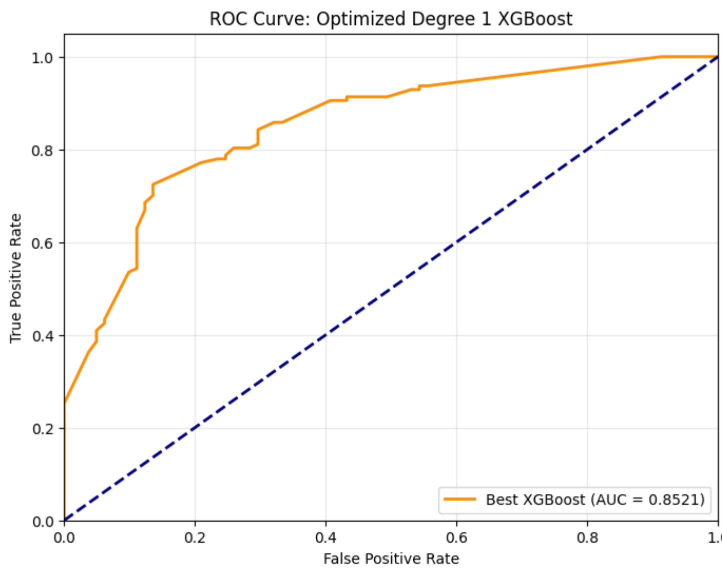

Figure 8. XGBoost ROC Curve: Displays the performance of the optimized gradient-boosted model, which achieved a high AUC of 0.852, significantly above random prediction.

Compared to logistic regression, the boosted tree model achieved stronger predictive performance, suggesting that nonlinear interactions between task types and subject domains play a meaningful role in memorization behavior.

5.4.2 Feature Importance Analysis. Feature importance was first examined using the XGBoost gain metric, which measures how much each feature contributes to reducing the model's loss across all decision trees.

Figure 9 presents the top features ranked by their signed contribution toward predicting the memorized class.

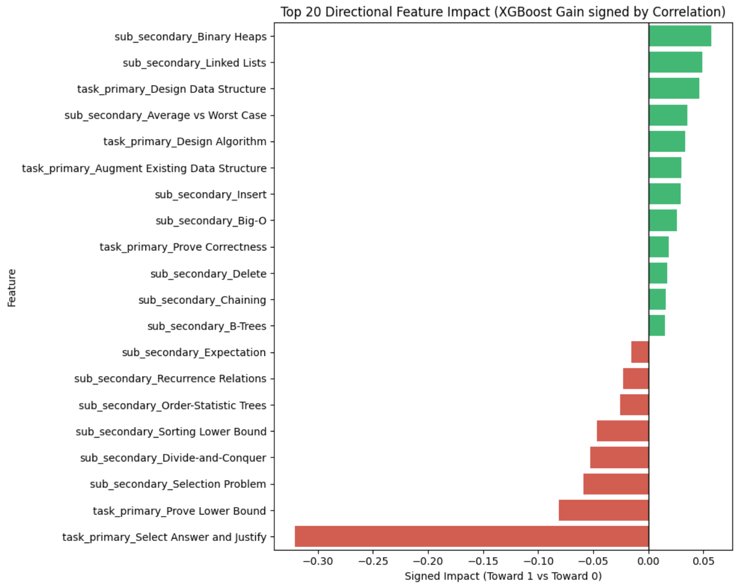

Figure 9. XGBoost Directional Feature Importance: Ranks features by their gain; "Binary Heaps" and "Linked Lists" are the top indicators for memorized retrieval, whereas "Lower Bound" proofs and "Select Answer and Justify" tasks are the strongest indicators of genuine reasoning.

Several interesting patterns emerge. Similarly to the logistic regression, topics such as binary heaps, linked lists, and general algorithm or data structure design tasks contribute positively toward predicting the memorized class.

In contrast, tasks involving justification, selection problems, divide-and-conquer analysis, or lower-bound proofs tend to contribute negatively toward the memorized classification - probably due to requirement of contextual reasoning or deeper structural arguments.

Similarly to the logistic regression, the strongest negative signal is associated with the task type `Select Answer and Justify`.

5.4.3 SHAP Analysis. To further interpret the model's behavior, we used SHAP (SHapley Additive exPlanations) values. SHAP provides a theoretically grounded method for decomposing the contribution of each feature to a model prediction.

For a given prediction $f(x)$, SHAP represents the model output as

$$f(x) = \phi_0 + \sum_{i=1}^{M} \phi_i$$

where ϕ_i represents the marginal contribution of feature i to the prediction. Figure 10 shows the SHAP summary plot for the most influential features.

The SHAP visualization confirms several insights suggested by the gain-based importance analysis. Features related to classical data structure topics such as heaps, linked lists, and algorithm design tend to push predictions toward the memorized class. Conversely, features associated with justification tasks, recurrence relations, or complex structural reasoning push predictions toward the non-memorized class.

Importantly, the SHAP plot also reveals the distribution of feature contributions across individual samples. This highlights that memorization is not determined by a single topic alone, but rather emerges from combinations of task types and conceptual domains.

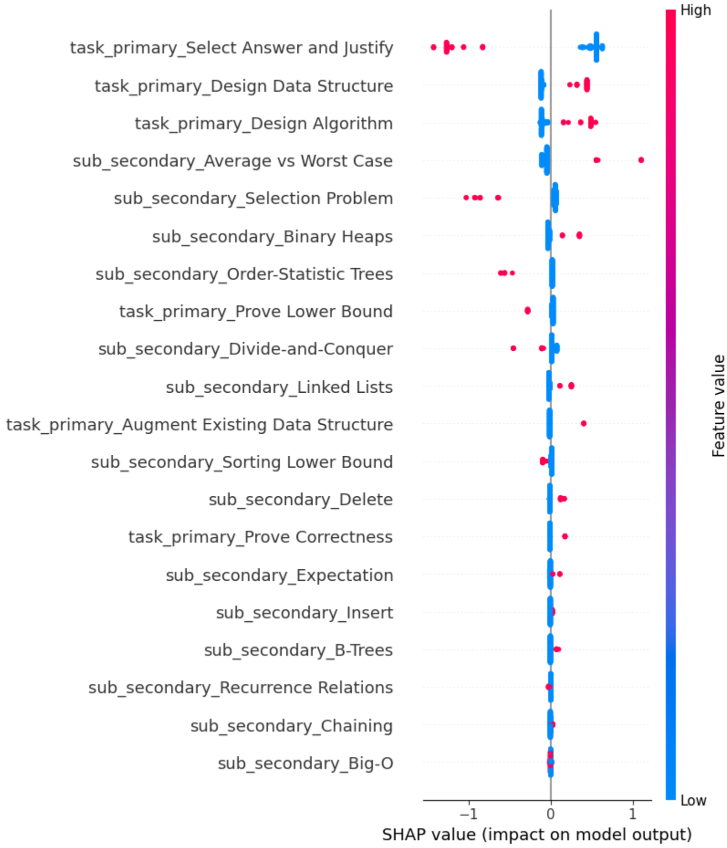

Figure 10. SHAP summary plot for the XGBoost classifier.

Overall, the XGBoost analysis reinforces the conclusion drawn from logistic regression: memorization behavior appears to be strongly associated with canonical algorithmic patterns and frequently taught topics, while reasoning-heavy or proof-oriented tasks are less likely to correspond to memorized outputs. The AUC improvement suggests that there are nonlinear interactions between some of the features.

5.5 Likely Memorized Grade Prediction (Attempt)

An additional analysis attempted to predict the grades obtained by questions classified as likely memorized. Since Phase 1 showed that the variance of grades across repeated solutions is significantly lower for likely memorized questions, the resulting grades are typically consistent for a given question. This raises the possibility that certain question characteristics might systematically lead to lower grades even within the memorized category. Identifying such factors could be useful, for example, in detecting topics where memorized knowledge is incomplete or unreliable and where targeted unlearning might be beneficial.

However, the distribution of grades within the likely memorized dataset is highly imbalanced. Among the $N = 127$ questions confidently classified as likely memorized, approximately 65% have an average grade of 90 or higher, while only about 13% have an average grade below 60.

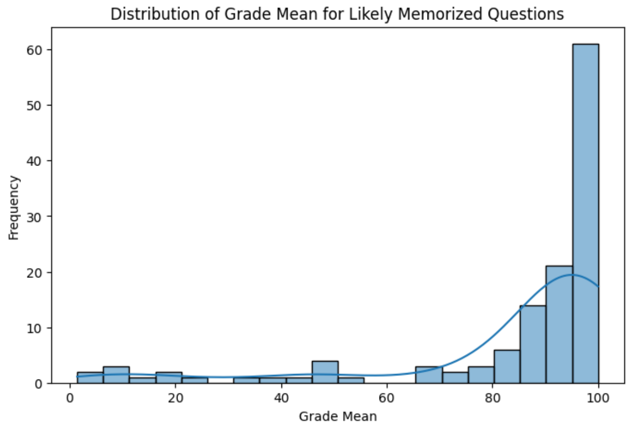

Figure 11. **Figure 11. Grade Distribution for Likely Memorized Questions:** This histogram illustrates that while the majority (65%) of questions confidently classified as likely memorized receive high average grades of 90 or above [cite: 563], a small subset (13%) consistently receives scores below 60. These low-performing outliers are concentrated in two distinct areas around grades of 10 and 50, demonstrating that the consistency of memorized patterns does not inherently guarantee the correctness of the solution.

Several modeling approaches were tested in order to predict the expected grade of a likely memorized question. Regression models included linear regression and XGBoost regression. In addition, classification models were tested by discretizing the grade distribution into categories (both binary and multi-class), using logistic regression and XGBoost classifiers. Because of the limited and imbalanced dataset, models were trained using weighted loss functions in order to balance precision and recall across classes. Evaluation was performed using k-fold cross-validation.

The results were not successful. Regression models produced very low and often negative adjusted R^2 values, indicating that the features used in the model explain little to none of the variance in the grades. Classification models showed some ability to fit the training data, but their validation performance (measured by AUC and F1 scores under cross-validation) was close to random prediction.

This outcome is likely due to the limited amount of available data and, in particular, the strong imbalance in the grade distribution. To further investigate whether specific topics systematically lead to lower grades, we performed a simple feature-level analysis. Questions were divided into two groups based on their mean grade, using a threshold of 70. For each task-type or subject label (encoded as one-hot features), we examined only the subset of questions where that feature was present and computed the proportion of questions in each class.

Table 5 presents the feature distribution over the classes of higher and lower grades among the likely memorized questions. The results suggest that the available feature set does not strongly explain grade variation among likely memorized questions. Although some topics appear slightly more frequently among lower-scoring questions, the differences are generally modest and often based on small sample counts. This further suggests that while memorization patterns can be detected through output variance and structural features, predicting the quality of the resulting solutions likely depends on additional factors not captured in the current representation of the questions.

Feature	Count	Class 0 (≤ 70)	Class 1 (>70)
sub_secondary_Heap Sort	2	0.50	0.50
sub_secondary_Dynamic Arrays	2	0.50	0.50
sub_secondary_Big-Θ	5	0.40	0.60
sub_secondary_Accounting Method	3	0.33	0.67
sub_secondary_Median-of-Medians	3	0.33	0.67
sub_secondary_AVL Trees	13	0.31	0.69
sub_secondary_Sorting Lower Bound	10	0.30	0.70
sub_secondary_Logarithms	14	0.29	0.71
sub_secondary_Arrays	30	0.27	0.73

Table 5. Feature Distribution by Grade Class: Demonstrates that specific topics or task-types do not strongly correlate with grade quality within the memorized group ($N = 127$), suggesting that model errors in memorized content are influenced by factors beyond the topic or task-type alone.

5.6 Reasoning and Performance Correlation

Attempts to identify correlations between model performance and student-related statistics were largely unsuccessful, mainly due to the limited availability of student data. The subset of questions containing student statistics was relatively small, which made it difficult to detect stable relationships between these variables and the model's performance across the different question groups (likely memorized, unlikely memorized, or the combined dataset). However, a more informative signal emerged when examining the relationship between the model's performance and its use of reasoning tokens.

First, we examined the relationship between the average number of reasoning tokens generated for a question and the model's average performance on that question. The examination yielded similar results on both likely memorized and unlikely memorized group, and on the combined groups. The results indicate a moderate negative correlation:

- Pearson correlation: $r = -0.3262$
- Spearman correlation: $\rho = -0.4015$
- p-value < 0.001
- $R^2 = 0.1064$
- Adjusted $R^2 = 0.0977$
- Linear model: $y = -0.0046x + 83.3452$

Figure 12 (left) illustrates this relationship. The negative slope indicates that questions for which the model generates longer reasoning chains tend to receive lower average grades. One possible interpretation is that problems requiring extended reasoning correspond to cases where the model lacks strong internal representations or memorized solution patterns. In such situations, the model attempts to construct a solution through longer reasoning chains, increasing the probability of logical errors or hallucinated intermediate steps. Conversely, questions that the model effectively “knows”—potentially due to memorization or strong pattern familiarity—tend to be solved with shorter reasoning traces and higher accuracy.

Second, we examined the relationship between the variability of reasoning tokens and the variability of model performance across multiple attempts for the same question. This analysis revealed a positive correlation:

- Pearson correlation: $r = 0.3600$
- Spearman correlation: $\rho = 0.4603$
- p-value < 0.001
- $R^2 = 0.1296$
- Adjusted $R^2 = 0.1211$
- Linear model: $y = 0.0046x + 10.6821$

As shown in Figure 12 (right), higher variance in reasoning tokens corresponds to higher variance in the model's performance. This suggests that reasoning token variance may serve as an indicator of task instability or ambiguity.

Despite the moderate Adjusted R^2 values (approximately 0.1), the very small p-values indicate that these relationships are statistically significant. Taken together, these results suggest that reasoning tokens provide a measurable proxy for the model's internal difficulty and confidence: questions that are solved quickly with short reasoning traces tend to produce higher and more stable grades, whereas questions requiring longer or inconsistent reasoning are associated with lower average performance and greater output variability. While the explained variance is moderate, this magnitude is still meaningful in the present setting, since model performance is plausibly affected by many latent and unobserved factors; accordingly, an Adjusted $R^2 \approx 0.1$ indicates a non-trivial explanatory signal.

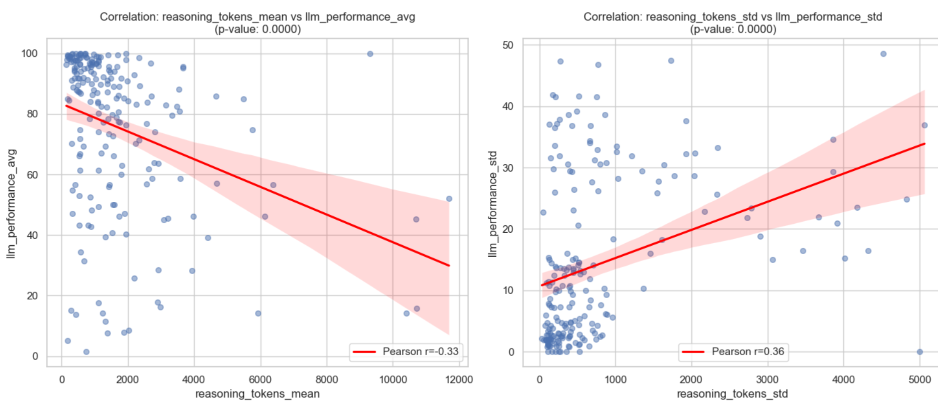

Figure 12. **Figure 12. Reasoning Token Usage vs. Model Performance:** Scatter plots showing a negative correlation ($\rho \approx -0.40$) between the length of reasoning and grade quality. This suggests that questions requiring longer reasoning chains tend to result in lower grades, potentially indicating a lack of relevant memorized content. In addition, a positive correlation ($\rho \approx 0.46$) between reasoning token variance and grade variance indicates that inconsistent reasoning paths serve as a proxy for task instability, leading to more unstable outcomes.

6 DISCUSSION

6.1 Limitations

Several limitations should be considered when interpreting the results of this study.

Validity and Completeness of Output Variance as a Memorization Estimator. While low output variance can arguably stem from inherent question simplicity rather than memorization, two factors support our estimator's validity. First, many questions in the "Likely Memorized" set involve complex proofs that are prevalent in Data Structures corpora but conceptually difficult, distinguishing memorization from "ease." Second, we identified a subset of low-variance questions where the model was consistently incorrect. This indicates that the model is reproducing a memorized error rather than benefiting from the task's simplicity, confirming that variance is not merely a proxy for difficulty. Finally, while output variance may not be a perfectly hermetic predictor of memorization, the high AUC scores demonstrate a strong correlation, confirming its robustness as an estimator.

Limited Dataset Size. A primary limitation is the relatively small size of the dataset. This constraint affected several aspects of the analysis.

First, it prevented the use of some analyses that could have been informative, particularly those involving statistics on student performance. Only about a quarter of the questions in the filtered dataset contained information about student grades, which limited the ability to investigate relationships between model behavior and human difficulty.

Second, the small dataset posed challenges for the predictive models in Phase 3. Many of the explanatory variables were encoded as one-hot features, and when interaction terms were included the number of features increased substantially. As a result, the models operated in a high-dimensional feature space relative to the number of observations, which can lead to instability and overfitting, a phenomenon commonly referred to as the curse of dimensionality.

Limited Control Over GPT-Based Pipeline Stages. Another limitation stems from the use of language models at multiple stages of the data pipeline.

An additional model was used to grade the generated solutions. In an earlier intermediate assignment this grading approach was validated against manual grading and showed strong agreement. Nevertheless, if more time and resources were available, a more extensive validation process would have been implemented for the dataset construction pipeline, including the labeling process, solution production for the likely memorized dataset and the automatic grading procedure.

Errors introduced by language models can broadly be divided into two categories: random noise and systematic bias. Random errors, such as occasional hallucinations, primarily reduce the statistical power of the analyses by adding noise to the measurements. In such cases, strong or statistically significant findings are likely to persist even if the noise were reduced.

Systematic biases are more concerning. These could arise, for example, if the model systematically overemphasizes certain aspects of the course material, ignores required elements in the solution analysis, or incorrectly attributes specific concepts during grading. Such biases, particularly in the generation and evaluation of solutions, could influence some of the observed results. With additional time, a more systematic audit of the pipeline would be conducted in order to identify and quantify possible bias sources.

Limited Set of Explanatory Variables for Memorization. The identification of likely memorized questions in earlier phases relied primarily on signals derived from output variance. While the results suggest a meaningful relationship between low variance and memorization-like behavior, this relationship is not necessarily unique. Memorization likely lies on a spectrum, and other factors may also produce low output variance.

If additional explanatory variables influencing memorization behavior were identified, some questions might be classified differently. This limitation also affects the predictive models developed in Phase 3 due to possible misclassifications.

Correclation vs. Causality. In the current framework, the one-hot labels describing task types and course topics represent independent variables, while the model's performance metrics serve as dependent variables. While it is plausible that some of these variables exert direct or indirect causal influence on memorization behavior, the analyses presented here cannot definitively establish causality.

Heuristic Methodology. Finally, it is important to note that several design choices throughout the pipeline were heuristic. In several cases, progress was driven by approaches that appeared to work well for the current dataset. for example, choice of metrics in the transition between phases 1 and 2, or using classification based on clustering in the transition between phases 2 and 3. While these choices produced coherent and statistically meaningful results, there is no guarantee that the same exact workflow would generalize to other datasets, domains or language models. However, due to the dependency between sequential phases, the value of each phase in this study can also be assessed by the extent to which it enabled meaningful results in the subsequent phases.

As a result, the pipeline presented here should be interpreted primarily as a proof of concept and guidelines rather than a finalized methodology. Future work could aim to formalize these decisions more rigorously and test their robustness across different datasets, tasks and other large language models.

6.2 Interpretation and Conclusions

The results of Phases 1 and 2 suggest a clear inverse relationship between output variance and the degree of memorization used by the model. Consistent with prior literature, lower variance across several dimensions—including solution grades, reasoning token usage, and semantic distance between generated answers—tends to indicate stronger reliance on memorized patterns. Importantly, this relationship was observed not only in short factual responses but also in complex, logical, and computational questions from the domain of Data Structures and Algorithms, whose answers are expressed as textual reasoning. Moreover, the results indicate that relatively simple variance-based metrics can serve as effective proxies for identifying memorization behavior.

The distribution of questions in the metric space suggests that memorization lies on a spectrum. However, the statistically significant clustering observed in Phase 2 indicates that this spectrum can still be meaningfully approximated by discrete categories, allowing questions to be classified as likely or unlikely memorized. The manual review supports this statement as well. However, the manual review could suggest a different classification for "borderline" questions that were classified as "likely memorized" in phase 2 - supporting the choice to filter the dataset in phase 3.

The cluster structure also provides insight into the behavior of non-memorized questions. Questions that are not strongly memorized tend to exhibit substantially higher variability across the measured metrics. In particular, both the variance of grades and the variance of reasoning token usage are considerably higher in these cases. This suggests that answers to such questions are less reliable overall.

Conversely, when a question appears to be memorized, the model tends to produce answers of consistent quality. However, consistency should not be mistaken for correctness: when a memorized answer is incorrect, the model will likely repeat the same mistake consistently. In practice, questions that appear strongly memorized tend to receive consistently high grades. Nevertheless, there are also cases in which memorized solutions consistently receive low grades, indicating that memorization alone cannot be relied upon as a predictor of solution quality.

In Phase 3, the predictive models demonstrated that the labels derived from course topics and task types contain meaningful information about memorization behavior. Although the resulting classifiers are far from perfect, they perform substantially better than random prediction. This indicates that certain structural properties of questions are associated with memorization, and can imply strong patterns the model was trained on. Analysis of the models also highlights the importance of feature interactions: in several cases, the effect of a feature when combined with others differs from its direct effect, suggesting that memorization behavior depends on combinations of topics and task formats rather than on individual concepts alone.

The attempt to predict grades within the likely memorized subset was largely unsuccessful. This failure is primarily explained by the highly imbalanced grade distribution and the lack of explanatory variables capable of distinguishing between consistently high and consistently low grades. As a result, the models were unable to identify stable predictors of grade variation within this subset.

Finally, the observed negative correlation between the average number of reasoning tokens and the model's average performance is particularly notable. While the relationship is statistically significant, it should not be interpreted causally. Several alternative explanations are possible. For instance, questions that require longer reasoning chains may simply be more difficult for the model, leading to lower accuracy regardless of the reasoning process itself. Nevertheless, the results suggest that additional reasoning does not reliably compensate for the challenges the model encounters when solving complex Data Structures problems.

Overall, examining the full pipeline reveals that it is possible to learn meaningful information about when a language model relies on memorization and what factors may contribute to this behavior. Although the results do not perfectly predict whether a specific answer is memorized or what causes memorization in every case, the findings across all three phases provide a proof of concept that such behavior can be studied using relatively simple metrics. Importantly, these metrics can be applied even to closed-source models such as ChatGPT, Gemini, or Claude, making the approach broadly applicable for analyzing memorization behavior in modern large language models.

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A APPENDIX

A.1 Code and Data Availability

All code used for data collection, preprocessing, analysis, and figure generation is publicly available at: https://github.com/liasoffer/LLMMemorization

In the repository root you could also find the stratified sample of questions that were reviewed manually.

A.2 Labels

A.2.1 Subject Label Taxonomy. The following taxonomy represents the hierarchical structure of topics used to label the specific concepts required to solve each Data Structures and Algorithms (DSA) question. For the purpose of the Phase 3 analysis, the first level of the hierarchy (the broad domains such as Data Structures or Algorithms) was excluded; the analysis focused exclusively on the second and third levels to identify more granular contributing factors.

- **Data Structures**
  - *Arrays & Lists:* Arrays, Dynamic Arrays, Circular Arrays, Linked Lists, Sentinel Nodes, Stacks, Queues, Deques.
  - *Trees:* Binary Search Trees (BST), Randomly Built BSTs, Tree Traversals, AVL Trees (Rotations, Join/Split, Balance Factor), Fibonacci Trees, Rank Trees (Select/Rank), Finger Trees, B-Trees (Node Splitting/Fusing, Top-Down/Bottom-Up).
  - *Heaps / Priority Queues:* Priority Queues, Binary Heaps, Binomial Heaps, Lazy Binomial Heaps, Fibonacci Heaps.
  - *Hashing:* Hash Tables, Direct Addressing, Chaining, Open Addressing (Linear Probing), Universal Hashing, Perfect Hashing, Modular Hash Functions.
  - *Disjoint Sets:* Union-Find, Union by Rank, Path Compression.
  - *Graphs:* Directed Graphs, Adjacency Matrix, Adjacency List.
- **Algorithms**
  - *Sorting:* Insertion Sort, Heap Sort, Quick Sort, Counting Sort, Radix Sort, Sorting Lower Bound, Topological Sort.
  - *Selection:* Selection Problem, Quick-Select, Randomized-Select, Median-of-Medians.
  - *Tree Operations:* Search, Insert, Delete, Successor, Predecessor, Min, Max.
  - *Union-Find Ops:* Make-Set, Find, Union, Link.
- **Analysis**
  - *Asymptotic:* Big-O, Big-Omega, Big-Theta, Little-o, Little-omega.
  - *Amortized:* Aggregate Method, Accounting Method, Potential Method, De-amortization.
  - *Recurrences:* Recurrence Relations, Divide-and-Conquer, Master Theorem.
  - *Expected:* Average vs. Worst Case, Expected Time, Randomized Algorithms.
- **Math**
  - *Discrete Math:* Logarithms, Growth Rates, Harmonic Series, Catalan Numbers, Fibonacci Numbers.
  - *Probability:* Random Variable, Indicator Variables, Expectation, Markov's Inequality.
  - *Recursion:* Induction, Recursive Definitions, Ackermann Function.

A.2.2 task-type attributes. These labels categorize the nature of the intellectual output required from the student.

Label ID	Description	Expected Output
DS_DESIGN	Construct a data structure with specific bounds.	Structure, Ops, Invariants
ALG_DESIGN	Design a deterministic algorithm.	Steps, Pseudocode
RANDOMIZED_ALG	Design an algorithm using randomness.	Algorithm, Expected Compl.
PROOF_CORRECT	Prove an algorithm or invariant is correct.	Formal/Informal Proof
COMPLEXITY_ANA	Analyze asymptotic time/space complexity.	Asymptotic Bound
EXACT_BOUND	Compute exact, non-asymptotic counts.	Math Expression, Integer
PROB_ANALYSIS	Analyze expected cases or probabilities.	Expected Value, Justification
AMORTIZED_ANA	Prove bounds via potential or accounting.	Amortized Bound
RECURRENCE	Solve recurrence for asymptotic bounds.	Closed-form Solution
ASYMPT_LAWS	Reason about function growth/Big-O laws.	Justification, Classification
LOWER_BOUND	Prove model-based impossibility results.	Lower Bound Argument
FEASIBILITY	Determine if a complexity is achievable.	Yes/No, Counterexample
COUNTEREXAMPLE	Disprove a claim with a concrete case.	Specific Example
TRACE_SIM	Trace execution on a given input.	Final/Intermediate States
AUGMENT_DS	Extend an existing structure (e.g., AVL).	Augmentation, Invariants
MCQ_TF	Select correct option and justify.	Selection, Proof/Refutation

Table 6. task-type attributes and Expected Deliverables

A.3 Prompts

A.3.1 Translation Prompts. The following prompt was used to standardize the dataset into English with LaTex syntax for analysis:

Question Translation:

> You are an expert in translating mathematical questions from Hebrew to English LaTeX format.
> # Your task is to translate the following question to English while preserving its mathematical integrity and formatting it correctly in LaTeX.
> # Ensure that all mathematical expressions are accurately represented. DO NOT add any opening or closing sentence, simply deliver the translation: \n

Note: The Answer Translation prompt follows an identical structure adapted for response text.

A.3.2 Question Solver Suffix. To ensure concise and comparable outputs, the following constraint was appended to solver queries:

> Provide a clear and concise final answer WITHOUT reasoning threads or repitition of question description. USE 10 SENTENCES OR LESS in the final answer.

A.3.3 Question Grader Prefix. The grading agent was initialized with the following persona and instructions:

> You are a strict, unbiased grader for a university-level data structures course.
> You will be given: 1. A student's answer to a question. 2. The official solution.
> **Your task:**
> - Carefully compare the student's answer to the official solution.
> - Check whether the student's reasoning, explanations, terminology, and conclusions are correct.
> - Evaluate accuracy, completeness, clarity, and correctness of definitions, algorithms, and complexity analysis.
> - Provide a grade from 0 to 100, where 100 is a perfect answer and 0 is completely incorrect.
>
> **Instructions:** 1. Identify the main components required in the official solution. 2. Check which components appear in the student answer. 3. Note any incorrect statements, missing details, or misunderstandings. 4. Assign a grade from 0 to 100 with a short justification.
> **OUTPUT FORMAT:**
> Grade: <number from 0–100>
> Justification: <your explanation>

A.3.4 Full Solution Generator. To produce gold-standard references, the following prompt was used:

> You are an expert instructor for a university-level data structures and algorithms course.
> You will be given a data structures exam question.
> **Your task:** Produce a complete, rigorous model solution that would receive 100/100 on a strict university exam. Assume the answer is graded for correctness, clarity, completeness, and proper use of terminology. Use standard definitions, accepted proofs, and well-known arguments. Present reasoning step by step, with clear logical flow. Include mathematical notation, recurrences, invariants, or proofs when appropriate. Clearly state assumptions and handle edge cases if relevant. Keep the explanation concise but fully sufficient for full credit.
> **Guidelines:**
> - Do not add unnecessary background or informal commentary.
> - Do not include grading rubrics or references to evaluation criteria.
> - Do not mention alternative incorrect approaches or reference model behavior.
> - Write as if this is the official solution distributed after an exam.
>
> **Formatting:** Use clear paragraphs and mathematical expressions. If the question has multiple parts, address each part explicitly. Avoid bullet points unless they improve clarity.
> **Output:** Return only the solution text.

A.3.5 *Classification and Labeling Prompts.* **Subject Labeling**

User Prompt:

> TASK: You are given a SOLUTION to a Data Structures or Algorithms exam question. Your job is to extract all SUBJECT TAGS corresponding to data structures, algorithmic techniques, or theoretical concepts that are ACTUALLY USED in the solution.
> **RULES:** Use ONLY subject attributes defined in the label taxonomy JSON. Do NOT include explanations. If no labels apply, return an empty array.
> **FORMAT:** Return EXACTLY the following JSON: `{ "subject_tags": [] }`

System Prompt:

> You are a deterministic classification engine for Data Structures and Algorithms exam content. You are given a fixed label taxonomy in JSON format. You must:
> • Use ONLY labels appearing in the taxonomy JSON.
> • Follow the output format exactly; output valid JSON only.
> • Produce no explanations, markdown, or extra text.
> ===== LABEL TAXONOMY (JSON) =====
> (...)

Task Type Labeling

User Prompt:

> TASK: You are given a DATA STRUCTURES or ALGORITHMS exam QUESTION. Your job is to identify what the student is required to do.
> **RULES:** Use ONLY task-type attributes defined in the label taxonomy JSON. Choose EXACTLY ONE `primary_task_type`. Choose `secondary_task_types` ONLY if explicitly required. Do NOT infer solution details.
> **FORMAT:** Return EXACTLY the following JSON:
> `{ "primary_task_type": "", "secondary_task_types": [] }`

System Prompt:

> You are a deterministic classification engine. (Instructions identical to Subject Labeling System Prompt).
> ===== LABEL TAXONOMY (JSON) =====
> (...)